

Mean Reversion

Betting Prices Return to "Normal" — Lesson Explainer

The Concept

Many prices and spreads oscillate around some underlying average rather than trending forever. Mean reversion strategies formalize 'this has gone too far' into a precise, repeatable rule based on statistical distance from that average.

The standard tool is the z-score: $(\text{current price} - \text{historical mean}) / \text{historical standard deviation}$. A z-score of +2 means the price is two standard deviations above its average - statistically unusual, and historically likely (though never certain) to pull back.

Pairs trading and statistical arbitrage extend this to the spread between two related assets, betting the gap itself reverts regardless of overall market direction.

Illustrative Example: Z-Score Signal Zones

A stock's 60-day historical mean is \$100, standard deviation \$5. Different prices translate into these z-scores and illustrative signals.

Current Price	Distance From Mean	Z-Score	Illustrative Signal
\$115	+\$15	+3.0	Strong Sell / Fade
\$110	+\$10	+2.0	Sell / Fade Threshold
\$101	+\$1	+0.2	No Signal - Near Mean
\$90	-\$10	-2.0	Buy / Fade Threshold
\$82	-\$18	-3.6	Strong Buy / Fade

A common rule enters at +2 (roughly the top/bottom 5% of a normal distribution) and exits as price reverts toward zero - illustrative only; real markets don't move in a neat bell curve.

5 Things to Know About Mean Reversion

1. Not every asset mean-reverts - a genuinely trending asset can extend rather than revert.
2. 'Historical mean' is a moving target - too short a window is noisy, too long is slow to adapt.
3. This is the same math as Bollinger Bands - a band touch is a z-score extreme, shown visually.
4. 'Catching a falling knife' is the classic failure - a stock cheap for real reasons can keep falling.
5. Pairs trading isolates the spread, not market direction - it can profit in both up and down markets.

4 Things to Check Before Trusting a Signal

- Choose your lookback window deliberately - test multiple windows, not just one.
- Ask why it's stretched - a fundamental change may mean it never reverts.
- Backtest with realistic costs - frequent, short-lived trades can be eaten by fees and slippage.
- Define an exit if it doesn't revert - a stop bounds the cost of a broken assumption.

Worth knowing:

this explains the statistical reasoning behind mean-reversion strategies using illustrative numbers
 - it isn't personalized financial advice, and no strategy here is a recommendation to trade. Past patterns are never guaranteed to repeat.

Quiz — Answer the questions, then check the key on the next page

1. What does a mean-reversion strategy assume?

- A. Prices always trend in one direction forever
- B. Trading volume predicts future prices
- C. A price that strays far from its historical average tends to pull back toward it
- D. Company earnings are irrelevant to price

2. How is a z-score calculated in this lesson?

- A. Current price divided by trading volume
- B. $(\text{Current price} - \text{historical mean}) / \text{historical standard deviation}$
- C. Current price multiplied by the P/E ratio
- D. Yesterday's closing price minus today's opening price

3. What is the classic failure mode of mean reversion described in this lesson?

- A. The strategy requires too much starting capital
- B. It only works on cryptocurrency markets
- C. It cannot be backtested at all
- D. 'Catching a falling knife' - a stock cheap for a genuine reason that keeps getting cheaper

4. What does pairs trading bet on, according to this lesson?

- A. That the spread between two historically-linked assets will revert, regardless of market direction
- B. That the entire stock market will rise
- C. That interest rates will fall
- D. That one specific company will go bankrupt

5. Why should mean-reversion signals be backtested with realistic trading costs?

- A. Trading costs don't affect mean-reversion strategies
- B. Regulators require it by law
- C. Mean-reversion trades are often frequent and short-lived, so costs can erase a paper edge
- D. It guarantees the strategy will be profitable

Answer Key

1. What does a mean-reversion strategy assume?

Correct answer: C. A price that strays far from its historical average tends to pull back toward it

Mean reversion assumes a stable underlying average exists and extreme deviations tend to correct over time.

2. How is a z-score calculated in this lesson?

Correct answer: B. $(\text{Current price} - \text{historical mean}) / \text{historical standard deviation}$

The z-score measures how many standard deviations the current price is from its historical mean.

3. What is the classic failure mode of mean reversion described in this lesson?

Correct answer: D. 'Catching a falling knife' - a stock cheap for a genuine reason that keeps getting cheaper

A statistically extreme z-score driven by real fundamental deterioration can keep extending past typical thresholds.

4. What does pairs trading bet on, according to this lesson?

Correct answer: A. That the spread between two historically-linked assets will revert, regardless of market direction

Pairs trading isolates the relationship between two related assets, aiming to profit regardless of overall market direction.

5. Why should mean-reversion signals be backtested with realistic trading costs?

Correct answer: C. Mean-reversion trades are often frequent and short-lived, so costs can erase a paper edge

Unrealistic cost assumptions in a backtest can make a losing strategy look profitable on paper.